score_forecasts

Li Shandross

3/25/2022

Load in truth data and forecasts

We load in the hospitalization truth data and our forecasts. (Note: The load forecast parentheses function wasn't working, so alternate code is written to achieve the same output.)

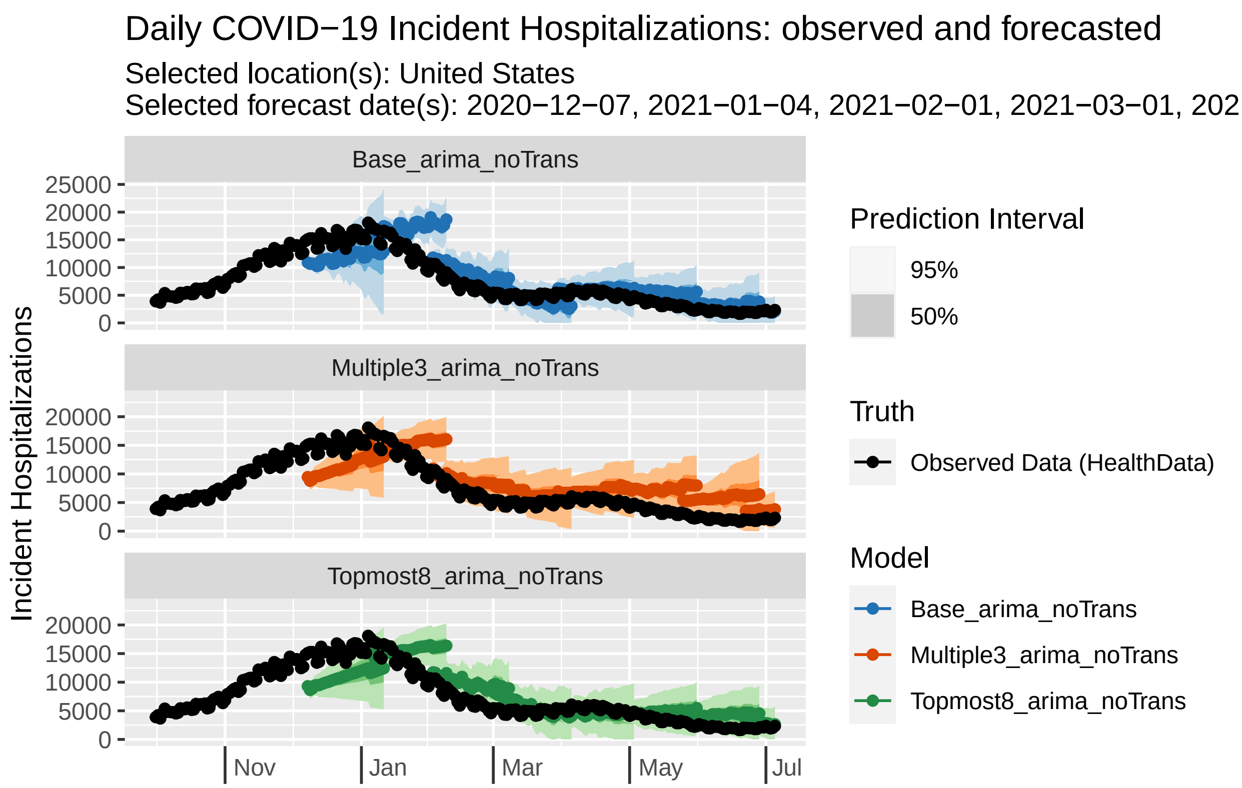
```
full_hosp_truth <- load_truth("HealthData",
                         "inc hosp",
                         temporal_resolution="daily",
                         data_location = "remote_hub_repo")

inc_hosp_targets <- paste(0:35, "day ahead inc hosp")

load("../data/loadedfc_scores.RData")


# models <- c("Base_arima_noTrans", "Multiple3_arima_noTrans", "Topmost8_arima_noTrans")
# file_paths <- list(list.files(path = "../data/Base_arima_noTrans/", pattern=".csv", full.names=TRUE),
#   list.files(path = "../data/covidTHieF-Multiple3_arima_noTrans/", pattern=".csv", full.names=TRUE),
#   list.files(path = "../data/Topmost8_arima_noTrans/", pattern=".csv", full.names=TRUE))
#
# forecasts <- read_csv(file_paths[[1]][1]) %>% mutate(model=models[1])
# for (i in 1:3) {
#   for (j in 1:31) {
#     if(i + j != 2) {
#       df <- read_csv(file_paths[[i]][j]) %>%
#         mutate(model = models[i])
#       forecasts <- rbind(forecasts, df)
#     }
#   }
# }
#
# forecasts2 <-
#   forecasts %>%
#     separate(target,
#       into=c("horizon", "temp"),
#       sep=" "
#     ) %>%
#     mutate(
#       horizon=as.numeric(horizon),
#       temporal_resolution="day",
#       target_variable="inc hosp"
```

```
#     ) %>%
#     select(model, forecast_date, location, horizon, temporal_resolution, target_variable, target_end_
#     left_join(hub_locations, by=c("location"="fips"))
```

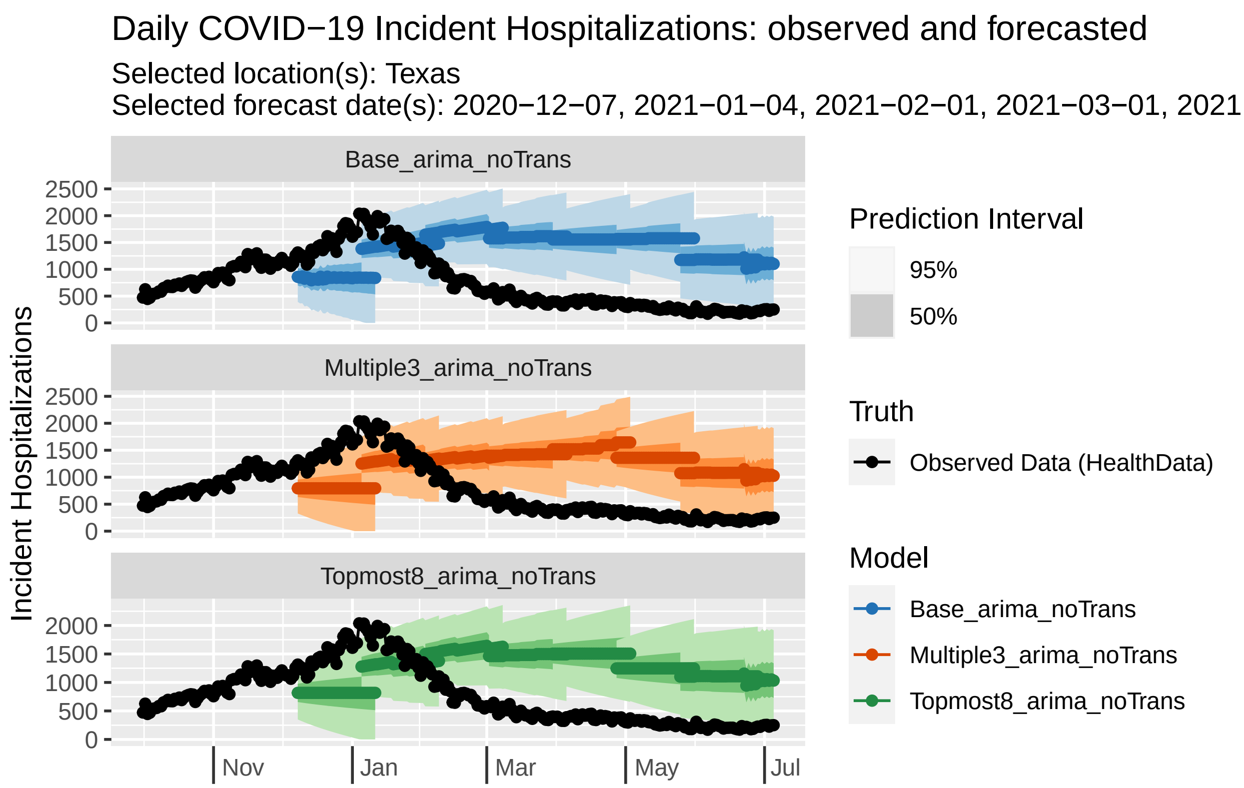
Plot Forecasts

We plot forecasts for three locations (Us National, Texas, and Vermont) for each of the three models. The two state locations were chosen as the highest and lowest hospitalization count locations. The plotted forecasts are spaced 4 weeks apart with 1 - 28 day ahead horizons, starting with the first forecast date, 12/07/20, until the end of the training period, 07/05/21.

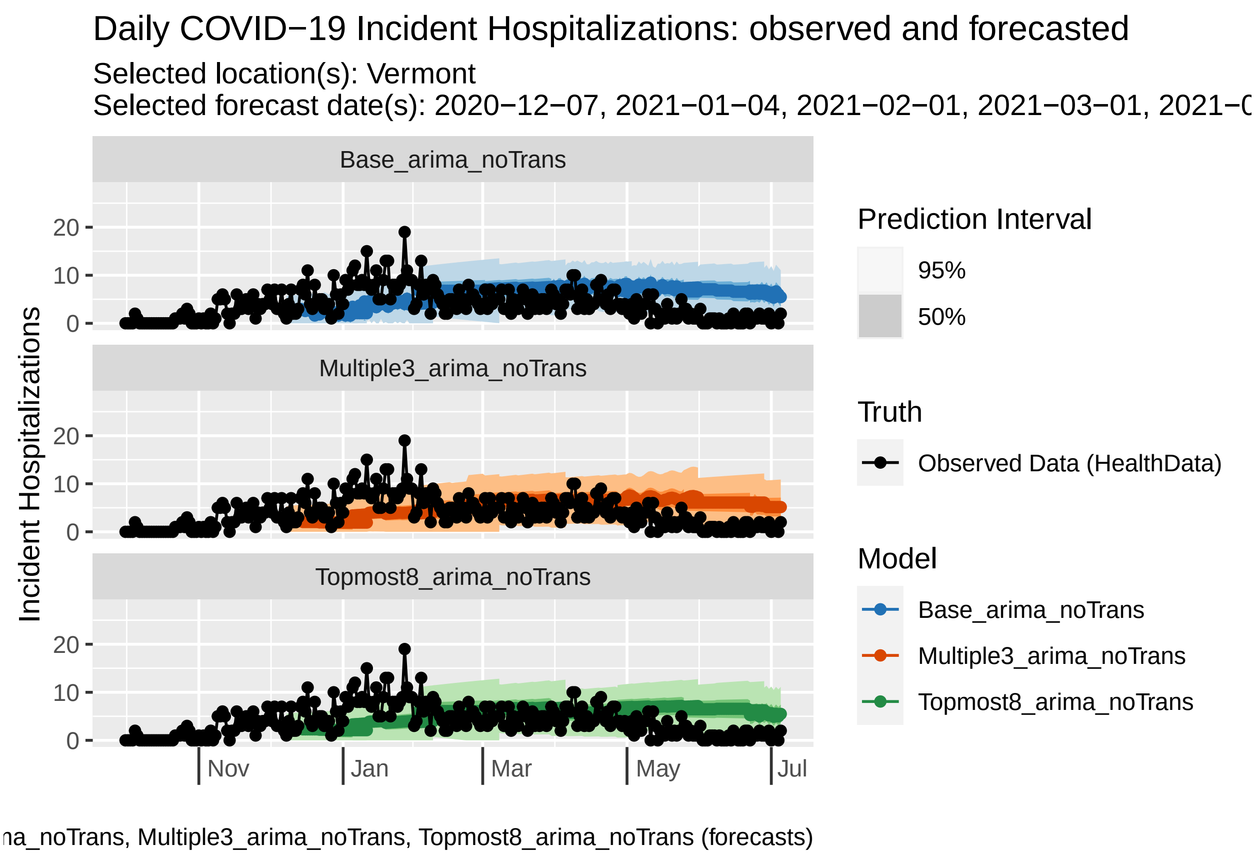
```
locs <- full_hosp_truth %>%
  filter(target_end_date <= as.Date("2021-07-05")) %>%
  group_by(location) %>%
  summarize(cum_value=sum(value)) %>%
  ungroup() %>%
  arrange(desc(cum_value)) %>%
  filter(row_number() %in% c(1, 2, 53)) %>%
  pull(location)

fc_dates <- c(as.Date("2020-12-07") + weeks(4*(0:7)))

for (i in 1:3) {
  fdat <-forecasts2 %>% filter(location == locs[i], forecast_date %in% fc_dates)

  p <- plot_forecasts(fdat,
                      target_variable = "inc hosp",
                      intervals = c(.5, .95),
                      truth_source = "HealthData",
                      use_median_as_point = TRUE,
                      facet = model ~.,
                      facet_nrow = 3,
                      facet_scales = "free_y",
                      fill_by_model = TRUE,
                      plot=FALSE)

  pt <- p +
    scale_x_date(name=NULL, limits=c(as.Date("2020-10-01"), as.Date("2021-07-05")),
             date_breaks = "2 months", date_labels = "%b") +
    theme(axis.ticks.length.x = unit(0.5, "cm"),
          axis.text.x = element_text(vjust = 7, hjust = -0.2))

  print(pt)
}
```



na_noTrans, Multiple3_arima_noTrans, Topmost8_arima_noTrans (forecasts)



na_noTrans, Multiple3_arima_noTrans, Topmost8_arima_noTrans (forecasts)



The plots don't show much visible difference between the three models (although this is partially due to the fact that the yscales are free and are not consistent even for the same location). However, it is clear that performance differs by location, all three models performing the worst for Texas and most closely following the trend line for the US National truth. We investigate model accuracy more deeply by scoring the forecasts for each model and calculating appropriate metrics.

Score forecasts

We calculate metrics for all three models, aggregating by horizon. Since hospitalizations are forecasted at a daily time scale, a new horizon week variable is created in which scoring metrics are averaged over from a single week Tuesday to the following Monday. We perform these calculations for the entire training phase as well as for each of the two waves that occurred during the training phase.

```
# scores <- score_forecasts(forecasts=forecasts2, return_format="wide", truth=full_hosp_truth, use_medi
#getwd()
# save(forecasts2, scores, file="../data/loadedfc_scores.RData")


# Overall metrics
overall_metrics <- scores %>%
  filter(horizon <= 28, target_end_date <= as.Date("2021-07-05")) %>%
  mutate(
    horizon_wk = case_when(
      horizon %in% c(1:7) ~ 1,
      horizon %in% c(8:14) ~ 2,
```

```
      horizon %in% c(15:21) ~ 3,
      horizon %in% c(22:28) ~ 4
    )
  ) %>%
  group_by(model, horizon_wk) %>%
  summarize(
    wis = mean(wis), mae=mean(abs_error),
    coverage_50 = mean(coverage_50),
    coverage_95 = mean(coverage_95)
  )
```

```
## 'summarise()' has grouped output by 'model'. You can override using the '.groups' argument.
```

```
knitr::kable(overall_metrics)
```

model	horizon_wk	wis	mae	coverage_50	coverage_95
Base_arima_noTrans	1	109.1251	158.3945	0.1834681	0.6361186
Base_arima_noTrans	2	117.5243	171.3417	0.1829166	0.6382563
Base_arima_noTrans	3	126.4471	185.6058	0.1829996	0.6515210
Base_arima_noTrans	4	136.8805	202.9858	0.1757013	0.6671658
Multiple3_arima_noTrans	1	113.7382	165.9560	0.2116801	0.6953279
Multiple3_arima_noTrans	2	114.0390	173.4713	0.2118227	0.7150293
Multiple3_arima_noTrans	3	117.6849	181.8501	0.2112052	0.7334424
Multiple3_arima_noTrans	4	129.3805	199.8321	0.2054507	0.7454328
Topmost8_arima_noTrans	1	101.7527	150.8079	0.2089847	0.7258760
Topmost8_arima_noTrans	2	107.2586	162.9258	0.2084766	0.7390092
Topmost8_arima_noTrans	3	113.3678	174.7495	0.2090874	0.7527917
Topmost8_arima_noTrans	4	122.0800	187.2452	0.2086453	0.7615054

```
# Metrics by wave
wave_metrics <- scores %>%
  filter(horizon <= 28, target_end_date <= as.Date("2021-07-05")) %>%
  mutate(
    horizon_wk = case_when(
      horizon %in% c(1:7) ~ 1,
      horizon %in% c(8:14) ~ 2,
      horizon %in% c(15:21) ~ 3,
      horizon %in% c(22:28) ~ 4
    ),
    wave = case_when(
      forecast_date <= as.Date("2021-03-17") ~"winter21",
      forecast_date >= as.Date("2021-03-17") ~"alpha"
    )
  ) %>%
  group_by(model, wave, horizon_wk) %>%
  summarize(
    wis = mean(wis), mae=mean(abs_error),
    coverage_50 = mean(coverage_50),
    coverage_95 = mean(coverage_95)
  )
```

```
## ‘summarise()‘ has grouped output by ’model’, ’wave’. You can override using the ‘.groups‘ argument.
```

```
knitr::kable(wave_metrics)
```

model	wave	horizon_wk	wis	mae	coverage_50	coverage_95
Base_arima_noTrans	alpha	1	122.07385	176.3052	0.1026056	0.5527403
Base_arima_noTrans	alpha	2	128.73489	186.1098	0.1101271	0.5490951
Base_arima_noTrans	alpha	3	135.30577	197.0310	0.1154883	0.5589882
Base_arima_noTrans	alpha	4	143.98265	212.2363	0.1109614	0.5662624
Base_arima_noTrans	winter21	1	96.17636	140.4837	0.2643306	0.7194969
Base_arima_noTrans	winter21	2	107.06115	157.5580	0.2508535	0.7214735
Base_arima_noTrans	winter21	3	118.76967	175.7040	0.2415094	0.7317161
Base_arima_noTrans	winter21	4	131.19870	195.5854	0.2274933	0.7478886
Multiple3_arima_noTrans	alpha	1	127.96210	188.1829	0.1185984	0.6323450
Multiple3_arima_noTrans	alpha	2	130.05442	197.9135	0.1264921	0.6440123
Multiple3_arima_noTrans	alpha	3	137.47929	211.2093	0.1368443	0.6547792
Multiple3_arima_noTrans	alpha	4	152.40491	234.6033	0.1331986	0.6554358
Multiple3_arima_noTrans	winter21	1	99.51420	143.7292	0.3047619	0.7583109
Multiple3_arima_noTrans	winter21	2	99.09123	150.6585	0.2914645	0.7813118
Multiple3_arima_noTrans	winter21	3	100.52980	156.4054	0.2756514	0.8016173
Multiple3_arima_noTrans	winter21	4	110.96105	172.0150	0.2632525	0.8174304
Topmost8_arima_noTrans	alpha	1	105.42445	159.2278	0.1439353	0.6837376
Topmost8_arima_noTrans	alpha	2	110.76658	168.6979	0.1515210	0.6923373
Topmost8_arima_noTrans	alpha	3	116.61312	179.2906	0.1582003	0.6976985
Topmost8_arima_noTrans	alpha	4	123.32035	188.9376	0.1570081	0.7003594
Topmost8_arima_noTrans	winter21	1	98.08089	142.3881	0.2740341	0.7680144
Topmost8_arima_noTrans	winter21	2	103.98454	157.5385	0.2616352	0.7825696
Topmost8_arima_noTrans	winter21	3	110.55521	170.8139	0.2531896	0.8005391
Topmost8_arima_noTrans	winter21	4	121.08781	185.8913	0.2499551	0.8104223

Based on the above metrics, we can see that generally the Topmost8_arima_noTrans model performed the best both in terms of WIS and MAE as well as PI Coverage, whereas the Multiple3_arima_noTrans model performed next best for PI Coverage but the Base_arima_noTrans model performed second best for WIS and MAE. There is not much difference between the metrics calculated for the entire training phase versus those split by wave, although notably the models perform better during the longer winter21 wave compared to the shorter alpha wave. The PI coverage is not great for any of the models, although it is slightly better for the 95% interval compared to the 50% one.